WORKED SOLUTION

STEP I 2009, Question 10

Pure • Wedge

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Read the question before you go on. It is in the STEP Support Programme's question database:

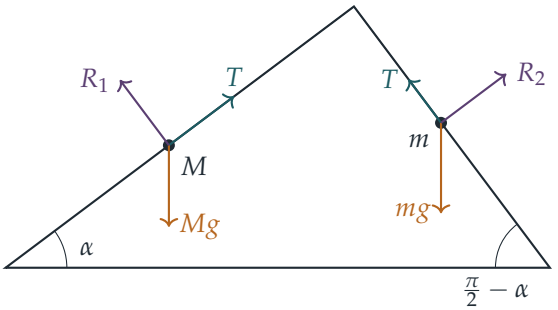
step.maths.org/questions/09-s1-q10

The key idea

Because the string is inextensible and the pulley smooth, the two particles move as one system: when M goes down its face a distance s, m goes up its face the same distance s, and the tension is the same on both sides. So we compare the two components of gravity along the faces, and Newton's second law for each particle gives the common acceleration. The last part is then an ordinary calculus maximisation in the single variable α.

Solution

We model the particles as point masses on smooth faces, with a light inextensible string over a smooth pulley, and assume the system starts from rest. Let T be the tension and R_1, R_2 the normal reactions on M and m.



The angle at the apex is $\pi - \alpha - (\frac{\pi}{2} - \alpha) = \frac{\pi}{2}$, so the faces are perpendicular. Gravity's component down the face of inclination α is $Mg\sin\alpha$, and down the face of inclination $\frac{\pi}{2} - \alpha$ it is $mg\sin(\frac{\pi}{2} - \alpha) = mg\cos\alpha$. The normal reactions are perpendicular to the faces, so they play no part in motion along them.

Part (i)

Take the direction "M down its face, m up its face" as positive, with acceleration a. Resolving along each face:

$$\begin{aligned} M: &\quad Mg\sin\alpha - T = Ma, \\ m: &\quad T - mg\cos\alpha = ma. \end{aligned}$$

Adding eliminates T:

$$a = \frac{g\,(M\sin\alpha - m\cos\alpha)}{M+m}.$$

Since $\cos\alpha > 0$ for $0 < \alpha < \frac{\pi}{2}$,

$$a > 0 \iff M\sin\alpha > m\cos\alpha \iff \tan\alpha > \frac{m}{M}.$$

So if $\tan\alpha > \frac{m}{M}$, the system, starting from rest, accelerates in the positive direction: M slides down its face, as required.

Ask yourself How do we know the string stays taut, so that these equations are the right ones?

From the second equation, $T = m(a + g\cos\alpha)$, which is positive when $a > 0$, so a taut string is consistent. Also, the string couldn't go slack: if it did, each particle would slide *down* its own face, moving both of them away from the pulley, which would need a longer string.

Part (ii)

Now $\tan\alpha = \frac{2m}{M}$, so $m = \frac{1}{2}M\tan\alpha$, and $\tan\alpha > \frac{m}{M}$, so by part (i) the motion is again with M moving down. Substituting $m = \frac{1}{2}M\tan\alpha$ and cancelling M:

$$\begin{aligned} a &= \frac{g\,(M\sin\alpha - \frac{1}{2}M\tan\alpha\cos\alpha)}{M + \frac{1}{2}M\tan\alpha} = \frac{g\,(\sin\alpha - \frac{1}{2}\sin\alpha)}{1 + \frac{1}{2}\tan\alpha} = \frac{\frac{1}{2}g\sin\alpha}{\frac{1}{2}(2+\tan\alpha)} \\ &= \frac{g\sin\alpha}{\tan\alpha + 2}, \end{aligned}$$

using $\tan\alpha\cos\alpha = \sin\alpha$. That is the required magnitude of the acceleration.

Maximising. The ratio m/M fixes α through $\tan\alpha = 2m/M$, so we maximise

$$f(\alpha) = \frac{\sin\alpha}{\tan\alpha + 2}, \qquad 0 < \alpha < \tfrac{\pi}{2}.$$

By the quotient rule, with $\frac{\mathrm{d}}{\mathrm{d}\alpha}\tan\alpha = \sec^2\alpha$,

$$f'(\alpha) = \frac{\cos\alpha\,(\tan\alpha + 2) - \sin\alpha\sec^2\alpha}{(\tan\alpha+2)^2} = \frac{\sin\alpha + 2\cos\alpha - \dfrac{\sin\alpha}{\cos^2\alpha}}{(\tan\alpha+2)^2}.$$

Multiplying the numerator by $\cos^2\alpha > 0$ and using $1 - \cos^2\alpha = \sin^2\alpha$:

$$\cos^2\alpha\,(\sin\alpha + 2\cos\alpha) - \sin\alpha = 2\cos^3\alpha - \sin\alpha\,(1 - \cos^2\alpha) = 2\cos^3\alpha - \sin^3\alpha.$$

So $f'(\alpha)$ has the same sign as $2\cos^3\alpha - \sin^3\alpha = \cos^3\alpha\,(2 - \tan^3\alpha)$. As $\tan\alpha$ increases from 0 to ∞, this is positive while $\tan^3\alpha < 2$ and negative once $\tan^3\alpha > 2$. Hence f increases and then decreases, and its maximum is at the unique stationary point

$$\tan^3\alpha = 2.$$

(You can also see this is a maximum because $f > 0$ on the interval while $f \to 0$ at both ends.) With $\tan\alpha = \frac{2m}{M}$ this says

$$\frac{8m^3}{M^3} = 2, \qquad \text{that is,} \qquad 4m^3 = M^3,$$

as required.

Checks

- Limiting cases of part (i): with $m = 0$ we get $a = g\sin\alpha$, a lone particle on a smooth slope; and when $\tan\alpha = m/M$ we get $a = 0$, the balance point.
- At the optimum, $\tan\alpha = 2^{1/3} \approx 1.26$, so $\alpha \approx 51.6^\circ$ and $m/M = 4^{-1/3} \approx 0.63$. A grid search of f over $(0, \frac{\pi}{2})$ put its maximum at the same α, with value $f \approx 0.240$, so the greatest acceleration is about $0.240g$.
- The general formula in part (i), with random masses and angles, agreed with the formula in part (ii) whenever $\tan\alpha = 2m/M$, and both equations of motion gave the same, positive, tension.

Common slips

- Using $mg\sin\alpha$ for m: its face is inclined at $\frac{\pi}{2} - \alpha$, so the component is $mg\cos\alpha$.
- Maximising with m held fixed and M varying (or the other way round) without noticing that a depends only on α. Writing everything in terms of α makes it a one-variable problem.

Beyond STEP

Systems like this, with bodies linked by constraints, are the starting point for Lagrangian mechanics: you pick one coordinate (here, the distance s moved along the string), and the constraint forces such as the tension and the normal reactions drop out automatically. A first-year course turns this into a systematic method that works for pulleys, pendulums and much more. For a clear treatment of Newtonian mechanics at this level, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

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